

$$\begin{aligned} 8x + 6y &= 34 & \dots(iii) \\ \text{and } 21x - 6y &= 24 & \dots(iv) \end{aligned}$$

On adding Eqs. (iii) and (iv), we get

$$\begin{aligned} (8x + 6y) + (21x - 6y) &= 34 + 24 \\ \Rightarrow 29x &= 58 \\ \therefore x &= 2 \end{aligned}$$

On putting $x = 2$ in Eq. (i), we get

$$8 + 3y = 17 \Rightarrow y = 3$$

Hence, $x = 2$ and $y = 3$ is the required solution.

21. (d) Given, $\sqrt{2}x - \sqrt{3}y = 0$

and $\sqrt{7}x + \sqrt{2}y = 0$

Here, $a_1 = \sqrt{2}$, $b_1 = -\sqrt{3}$

$a_2 = \sqrt{7}$ and $b_2 = \sqrt{2}$

As the equations are homogeneous equations and also

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. So, equation has one solution,

$\therefore x = y = 0 \Rightarrow x + y = 0$

Hence, the value of $x + y$ is zero.

22. (a) Let the starting salary be ₹ x and annual increment be ₹ y .

By given condition, $x + 4y = 1500$... (i)

and $x + 10y = 1800$... (ii)

On subtracting Eq. (i) from Eq. (ii), we get

$$6y = 300 \Rightarrow y = 50$$

On putting $y = 50$ in Eq. (i), we get

$$x + 200 = 1500$$

$\therefore x = ₹ 1300$

23. (b) Put $\frac{1}{x+y} = A$ and $\frac{1}{x-y} = B$

$\Rightarrow 44A + 30B = 10$... (i)

and $55A + 40B = 13$... (ii)

On multiplying Eq. (i) by 4 and Eq. (ii) by 3 and subtracting, we get

$$176A + 120B = 40$$

$$165A + 120B = 39$$

$$\frac{11A}{11A} = \frac{1}{1}$$

$\Rightarrow A = 1/11 \Rightarrow x + y = 11$... (iii)

On putting the value of A in Eq. (i), we get

$$4 + 30B = 10 \Rightarrow 30B = 6$$

$\therefore B = \frac{1}{5}$, $x - y = 5$... (iv)

On adding Eqs. (iii) and (iv), we get

$$2x = 16$$

$\therefore x = 8$

On putting $x = 8$ in Eq. (iii), we get

$$8 + y = 11 \Rightarrow y = 3$$

Hence, $x = 8$ and $y = 3$.

24. (c) Set I $2x - y = 0$ and $6x - 3y = 0$

Here, $a_1 = 2$, $b_1 = -1$

and $a_2 = 6$, $b_2 = -3$

$\Rightarrow \frac{a_1}{a_2} = \frac{2}{6} = \frac{1}{3} \Rightarrow \frac{b_1}{b_2} = \frac{-1}{-3} = \frac{1}{3}$

$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2}$

So, system of equations have infinitely many solutions.

Set II $3x - 4y = 0$ and $12x - 20y = 0$

Here, $a_1 = 3$, $b_1 = -4$

and $a_2 = 12$, $b_2 = -20$

$\therefore \frac{a_1}{a_2} = \frac{3}{12} = \frac{1}{4} \Rightarrow \frac{b_1}{b_2} = \frac{-4}{-20} = \frac{1}{5}$

Here $\frac{1}{4} \neq \frac{1}{5}$

So, system of equations has unique solution.

25. (a) Let the fraction be $\frac{x}{y}$.

Condition I $\frac{x+1}{y+1} = \frac{4}{5}$

$\Rightarrow 5(x+1) = 4(y+1)$

$\Rightarrow 5x - 4y = -1$... (i)

Condition II $\frac{x-5}{y-5} = \frac{1}{2}$

$\Rightarrow 2(x-5) = (y-5)$

$\Rightarrow 2x - y = 5$... (ii)

On multiplying Eq. (ii) by 4 and subtracting from Eq. (i), we get

$$5x - 4y = -1$$

$$8x - 4y = 20$$

$$\frac{-}{-3x} = \frac{-}{-21}$$

$\therefore x = 7$

Put the value of x in Eq. (i),

$\Rightarrow 35 - 4y = -1 \Rightarrow y = 9$

Hence, the fraction is $\frac{7}{9}$.

26. (a) Given equations are,

$3x + 4y = 6$... (i)

and $6x + 8y = k$... (ii)

Here, $a_1 = 3$, $b_1 = 4$, $c_1 = 6$

$a_2 = 6$, $b_2 = 8$, $c_2 = k$

Since the system of equations represents coincident line

$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, i.e. $\frac{3}{6} = \frac{4}{8} = \frac{6}{k}$

$\Rightarrow 4k = 6 \times 8 \Rightarrow k = \frac{48}{4} = 12$

Hence, the value of k is 12.

27. (c) Let the length of rectangle be x m and breadth be y m.

$\therefore$ Original area of rectangle = xy

Case I Length = $(x + 7)$ m and breadth = $(y - 3)$ m

$\therefore$ New area of rectangle = $(x + 7)(y - 3)$

Given, $xy = xy - 3x + 7y - 21$

$\Rightarrow 3x - 7y = -21$... (i)

Case II Length = $(x - 7)$

and Breadth = $(y + 5)$

$\therefore$ New area of rectangle = $(x - 7)(y + 5)$

Given, $xy = xy + 5x - 7y - 35$

$\Rightarrow 5x - 7y = 35$... (ii)

On subtracting Eq. (ii) from Eq. (i), we get

$$3x - 7y = -21$$

$$5x - 7y = 35$$

$$\frac{-}{-2x} = \frac{-}{56}$$

$\therefore x = 28$ m

On putting $x = 28$ in Eq. (ii), we get

$$5 \times 28 - 7y = 35 \Rightarrow -7y = 35 - 140$$

$\therefore y = 15$

$\therefore$ Area of rectangle = $xy = 15 \times 28 = 420 \text{ m}^2$

28. (b) Given, $2^{x+y} = \sqrt{8}$, $2^{x-y} = \sqrt{8}$

or $2^{x+y} = 2\sqrt{2}$, $2^{x-y} = 2\sqrt{2}$

or $2^{x+y} = 2^{3/2}$, $2^{x-y} = 2^{3/2}$

On comparing both sides, we get

$x + y = \frac{3}{2}$... (i)

and $x - y = \frac{3}{2}$... (ii)

On adding Eqs. (i) and (ii), we get $2x = 3$

$\Rightarrow x = \frac{3}{2}$

Hence, the value of x is $\frac{3}{2}$.

29. (c) Let A has x mangoes and B has y mangoes.

Case I $x + 30 = 2(y - 30)$

$\Rightarrow x + 30 = 2y - 60$

$x - 2y = -90$... (i)

Case II $(y + 10) = 3(x - 10)$

$y = 3x - 30 - 10$

$3x - y = 40$... (ii)

On solving Eqs. (i) and (ii), we get

$x = 34$, $y = 62$

So, A has 34 mangoes.

30. (c) Let number of students in room A be x and in room B be y .

So, by given condition

$$x - 10 = y + 10 \Rightarrow x - y = 20 \dots (i)$$

$$\text{and } x + 20 = 2(y - 20)$$

$$x - 2y = -60 \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(x - y) - (x - 2y) = 20 + 60 \Rightarrow y = 80$$

Hence, number of students in room B is 80.

31. (d) Let the number of passengers in the starting be x .

Number of passengers after first halt

$$= \left[x - \frac{x}{3} \right] + 120 = \frac{2}{3}x + 120$$

and number of passengers after second halt

$$= \frac{1}{2} \left[\frac{2}{3}x + 120 \right] + 100$$

But number of passengers after second halt = 240

$$\therefore \frac{1}{2} \left[\frac{2}{3}x + 120 \right] + 100 = 240$$

$$\Rightarrow \frac{2}{3}x + 120 = 280 \Rightarrow \frac{2}{3}x = 160$$

$$\Rightarrow x = 240$$

32. (c) Given system of equations are

$$x + 2y = 3 \text{ and } 3x + 6y = 9$$

$$\text{Here, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{3}$$

Hence, given system of the equation has infinitely many solutions.

33. (b) Let x be the ten's digit and y be the unit's digit of two-digit number.

$$\text{By given condition, } x + y = 8$$

$$\text{and } (10x + y) - (10y + x) = 18 \dots (i)$$

$$\Rightarrow 9x - 9y = 18 \Rightarrow x - y = 2 \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$2x = 10 \Rightarrow x = 5$$

On putting $x = 5$ in Eq. (i), we get

$$\Rightarrow y = 8 - 5 = 3 \Rightarrow x = 5 \text{ and } y = 3$$

$\therefore$ Required difference of digits,

$$x - y = 5 - 3 = 2$$

34. (a) Given, $(x, y) = (4, 1)$ and

$$mx + y = 2x + ny = 5$$

On putting $x = 4$ and $y = 1$, we get

$$\therefore m(4) + 1 = 2 \times 4 + n = 5$$

$$\therefore 4m + 1 = 5 \text{ and } 8 + n = 5$$

$$\Rightarrow n = -3 \text{ and } m = 1$$

$$\therefore m + n = 1 - 3 = -2$$

35. (d) Given, $2^a + 3^b = 17$

$$\text{and } 2^{a+2} - 3^{b+1} = 5$$

$$\Rightarrow 2^a \times 2^2 - 3^b \times 3^1 = 5$$

$$\Rightarrow 4 \cdot 2^a - 3 \cdot 3^b = 5$$

$$\text{Let } 2^a = x \text{ and } 3^b = y$$

$$\text{Then, } x + y = 17 \dots (i)$$

$$4x - 3y = 5 \dots (ii)$$

On multiplying Eq. (i) by 3 and adding to Eq. (ii), we get

$$3x + 3y = 51$$

$$4x - 3y = 5$$

$$7x = 56$$

$$\therefore x = 8$$

On putting the value of x in Eq. (i), we get

$$8 + y = 17 \Rightarrow y = 9$$

$$\text{Now, } 2^a = x \Rightarrow 2^a = 8 = (2)^3$$

$$\therefore a = 3 \text{ and } 3^b = y = 9 \Rightarrow 3^b = 3^2$$

$$\therefore b = 2$$

$$\text{Hence, } a = 3 \text{ and } b = 2$$

36. (b) Let smaller number be x .

$$\text{and larger number} = 80 - x$$

$$\text{By given condition, } 80 - x = 4x + 5$$

$$\Rightarrow 5x = 75 \Rightarrow x = 15$$

Hence, the smaller number is 15.

37. (c) Given equations are,

$$\frac{2}{x} + \frac{3}{y} = \frac{9}{xy} \Rightarrow 2y + 3x = 9 \dots (i)$$

$$\text{and } \frac{4}{x} + \frac{9}{y} = \frac{21}{xy} \Rightarrow 4y + 9x = 21 \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$x = 1 \text{ and } y = 3$$

$$\therefore x + y = 1 + 3 = 4$$

38. (b) Since, the equation $kx - y = 2$ and $6x - 2y = 3$ have a unique solution

$$\therefore \frac{k}{6} \neq \frac{1}{2} \Rightarrow k \neq 3$$

39. (b) Three lines $a_1x + b_1y + c_1 = 0$

$$a_2x + b_2y + c_2 = 0$$

$$\text{and } a_3x + b_3y + c_3 = 0$$

which are non-parallel and non-collinear they have only one solution if they meet in a common point in this case these lines are called concurrent lines.

40. (c) Let the unit's digit be x and ten's digit's be y . Then the two-digit number

$$= 10y + x$$

$$\text{By given condition, } x + y = 10 \dots (i)$$

$$\text{and } 10y + x - 18 = 10x + y$$

$$\Rightarrow 9x - 9y = -18 \Rightarrow x - y = -2$$

On solving Eqs. (i) and (ii), we get

$$x = 4 \text{ and } y = 6$$

$$\therefore \text{Required product} = xy = 4 \times 6 = 24$$

41. (d) Since, given system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\therefore \frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \Rightarrow k = 10$$

42. (a) Given, $x - y = 0.9 \dots (i)$

$$\text{and } 11(x + y)^{-1} = 2 \dots (ii)$$

$$\Rightarrow 2x + 2y = 11$$

On multiplying Eq. (i) by 2 and adding Eqs. (i) and (ii), we get

$$4x = 12.8 \Rightarrow x = 3.2$$

$$\text{From Eq. (i) } y = 3.2 - 0.9 = 2.3$$

43. (d) Let Pooja's initial salary be ₹ x and fixed increment every year be ₹ y .

$$\text{By given condition, } x + 3y = 4200 \dots (i)$$

$$\text{and } x + 8y = 6800 \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(x + 3y) - (x + 8y) = 4200 - 6800$$

$$\Rightarrow -5y = -2600 \Rightarrow y = ₹520$$

On putting $y = 520$ in Eq. (i), we get

$$x = ₹2640$$

44. (b) Let the fare of ticket from station P to station Q is ₹ x and that from station P to station R is ₹ y .

$$\text{By given condition, } x + y = 42$$

$$\text{and } 5x + 10y = 350$$

On solving Eqs. (i) and (ii), we get

$$x = 14 \text{ and } y = 28$$

Hence, fare from station P to station Q is ₹ 14.

45. (a) Let x represent school bags, y represent sweaters, z represent books. System of equations:

$$x + y + z = 200 \dots (i)$$

$$z = x + y \dots (ii)$$

$$20x + 25y + 5z = 2800 \dots (iii)$$

putting $z = x + y$ in Eq. (i)

$$(x + y) + x + y = 200, 2x + 2y = 200$$

$$x + y = 100, x = 100 - y$$

putting $x = 100 - y$ in Eq. (iii)

$$20(100 - y) + 25y + 5 = 2800$$

$$[2000 - 20y + 25y + 5] = 2800$$

$$2000 - 20y + 25y + 5 = 2800$$

$$5y = 300 \Rightarrow y = 60$$

$$x = 100 - 60 \Rightarrow x = 40$$

$$z = 40 + 60 \Rightarrow z = 100$$

40 school bags, 60 sweaters, 100 books

46. (b) Let the share of Rajesh be R , Sonal be S and Chetan be C , respectively.
According to question, $2R = 3S = 5C$
also, $6R + 6S = 150$
 $\Rightarrow \frac{3}{2} \times 6R + 6S = 150$
 $\Rightarrow 9S + 6S = 150 \Rightarrow S = 10$
 $\therefore C = \frac{3}{5}S = \frac{3}{5} \times 10 = ₹ 6$.

47. (a) $R + C + S = 155$

$$R + \frac{2}{5}R + \frac{2}{3}R = 155$$

$$\frac{15R + 6R + 10R}{15} = 155 \Rightarrow R = ₹ 75$$

48. (c) The graph of $ax + by = c$, $dx + ey = f$ will be coincident, if the system has infinite number of solutions.
So, statement II is false.

Thus, statements I and III are correct.

49. (c) Let the two numbers be x and y , respectively. According to the question,

$$x + y = 20 \quad \dots(i)$$

$$\text{and } xy = 75 \quad \dots(ii)$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{y+x}{xy} = \frac{20}{75} = \frac{4}{15}$$

[from Eqs. (i) and (ii)]

50. (b) Given equations are,

$$3^{x+y} = 81 \Rightarrow 3^{x+y} = 3^4$$

$$\Rightarrow x + y = 4 \quad \dots(i)$$

$$\text{and } 81^{x-y} = 3 \Rightarrow (3^4)^{x-y} = 3^1$$

$$\Rightarrow x - y = \frac{1}{4} \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$\Rightarrow 2x = \frac{17}{4} \Rightarrow x = \frac{17}{8}$$

51. (d) Given equations are,

$$\frac{a}{b} - \frac{b}{a} = \frac{x}{y} \quad \dots(i)$$

$$\text{and } \frac{a}{b} + \frac{b}{a} = x - y \quad \dots(ii)$$

From Eqs. (i) and (ii),

$$\frac{a}{b} + \frac{b}{a} = x - \left(\frac{\left(\frac{a}{b} - \frac{b}{a} \right)x - x}{\left(\frac{a}{b} - \frac{b}{a} \right)} \right) = \frac{a}{b} - \frac{b}{a}$$

$$\Rightarrow \left(\frac{a}{b} + \frac{b}{a} \right) \left(\frac{a}{b} - \frac{b}{a} \right) = x \left(\frac{a}{b} - \frac{b}{a} - 1 \right)$$

$$\Rightarrow \left(\frac{a^2}{b^2} - \frac{b^2}{a^2} \right) = x \left(\frac{a^2 - b^2 - ab}{ab} \right)$$

$$\Rightarrow x = \frac{ab}{(a^2 - b^2 - ab)} \times \left(\frac{a^4 - b^4}{a^2 b^2} \right)$$

$$\Rightarrow x = \frac{(a^4 - b^4)}{a^2 - b^2 - ab} \cdot \frac{1}{ab} \\ = \frac{(a-b)(a+b)(a^2 + b^2)}{ab(a^2 - b^2 - ab)}$$

52. (d) Given equations are,

$$3x + y = 4 \quad \dots(i)$$

$$6x + 2y = 8 \quad \dots(ii)$$

$$\text{Here, } a_1 = 3, b_1 = 1 \text{ and } c_1 = 4$$

$$\text{and } a_2 = 6, b_2 = 2 \text{ and } c_2 = 8$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2}$$

So, the system of equations has infinitely many solutions.

53. (c) Let the two numbers be x and y , respectively.

Given, sum of two numbers = 7

$$\Rightarrow x + y = 7 \quad \dots(i)$$

and sum of their squares = 25

$$\Rightarrow x^2 + y^2 = 25 \quad \dots(ii)$$

Now, we have

$$(x + y)^2 = (x^2 + y^2) + 2xy$$

$$\Rightarrow (x + y)^2 - (x^2 + y^2) = 2xy$$

$$\Rightarrow xy = \frac{1}{2}[(x + y)^2 - (x^2 + y^2)]$$

$$= \frac{1}{2}[(7)^2 - (25)]$$

[from Eqs. (i) and (ii)]

$$= \frac{1}{2}(49 - 25) = \frac{1}{2} \times 24 = 12$$

Hence, the product of the two numbers is 12.

Solutions (Q. Nos. 54-55) Let the digit in unit's place be y and that in ten's place be x .

Then, the two-digit number is given by $10x + y$.

Number obtain by reversing the order of digit = $10y + x$

Now, by given condition,

$$x + y = 10 \quad \dots(i)$$

$$\text{and } (x + 10y) + 36 = (y + 10x)$$

$$\Rightarrow -9y + 9x = 36$$

$$\Rightarrow x - y = 4 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$2x = 14, x = 7$$

On putting the value of x in Eq. (i), we get

$$7 + y = 10 \Rightarrow y = 3$$

54. (b) The required number is 73.

So, the number is a multiple of a prime number.

55. (a) Required product of two digits

$$= 3 \times 7 = 21$$

56. (a) Let the cost of one chair and one table be ₹ x and ₹ y , respectively.

By given condition,

$$10x + 6y = 6200 \quad \dots(i)$$

$$\text{and } 3x + 2y = 1900$$

$$\Rightarrow 9x + 6y = 5700 \quad \dots(ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(10x + 6y) - (9x + 6y) = 6200 - 5700$$

$$\Rightarrow x = 500$$

$$\therefore x = ₹ 500$$

From Eq. (i), $5000 + 6y = 6200$

$$\Rightarrow 6y = 1200 \Rightarrow y = ₹ 200$$

$\therefore$ Cost of 4 chairs and 5 tables

$$= 4x + 5y = 4 \times 500 + 5 \times 200$$

$$= 2000 + 1000 = ₹ 3000$$

57. (b) Let bus starts with x number of passengers.

After 1st stoppage, number of passengers

$$= x - \frac{x}{5} + 40 = \frac{5x - x + 200}{5}$$

$$= \frac{4x + 200}{5}$$

After 2nd stoppage, number of passengers

$$\frac{4x + 200}{5} - \frac{4x + 200}{5 \times 2} + 30$$

According to question,

$$\Rightarrow \frac{4x + 200}{5} - \frac{4x + 200}{10} + 30 = 70$$

$$\frac{4x + 200}{5} \left(1 - \frac{1}{2} \right) + 30 = 70$$

$$\Rightarrow \frac{4x + 200}{5} \times \frac{1}{2} = 40$$

$$\Rightarrow 4x + 200 = 400 \Rightarrow 4x = 200$$

$$\therefore x = \frac{200}{4} = 50$$

Hence, the number of passengers is 50.

58. (d) We have, $x + y - 7 = 0$

$$\Rightarrow x + y = 7 \quad \dots(i)$$

$$\text{and } 3x + y - 13 = 0$$

$$\Rightarrow 3x + y = 13 \quad \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$3x + y = 13$$

$$x + y = 7$$

$$\underline{- \quad - \quad -}$$

$$2x = 6$$

$$\therefore x = 3$$

On putting the value of x in Eq. (i), we get

$$3 + y = 7 \Rightarrow y = 4$$

$$\text{Now, } 4x^2 + y^2 + 4xy = (2x + y)^2$$

$$= (2 \times 3 + 4)^2$$

$$= (6 + 4)^2$$

$$= 10^2 = 100$$

59. (b) Given, $\frac{x}{2} + \frac{y}{3} = 4 \Rightarrow \frac{3x + 2y}{6} = 4$

$$\Rightarrow 3x + 2y = 24 \quad \dots(i)$$

$$\text{and } \frac{2}{x} + \frac{3}{y} = 1 \Rightarrow \frac{2y + 3x}{xy} = 1$$

$$\Rightarrow 2y + 3x = xy \quad \dots(ii)$$

On comparing Eqs. (i) and (ii), we get $xy = 24$

There are 8 possibilities for x and y , respectively.

$$\begin{array}{ll} 1 \times 24 = 24 & 6 \times 4 = 24 \\ 2 \times 12 = 24, & 8 \times 3 = 24 \\ 3 \times 8 = 24, & 12 \times 2 = 24 \\ 4 \times 6 = 24, & 24 \times 1 = 24 \end{array}$$

On putting all the values of x and y in given equations we get that only $x = 4$ and $y = 6$ satisfy the equations.

$$\therefore x + y = 4 + 6 = 10$$

60. (a) Given equations are,

$$\begin{array}{ll} x + y = 5 & \dots(i) \\ y + z = 10 & \dots(ii) \\ z + x = 15 & \dots(iii) \end{array}$$

from Eq. (i), we have $y = 5 - x$

put the value of x in Eq. (ii),

$$\begin{array}{ll} z + 5 - x = 10 & \\ \Rightarrow z - x = 5 & \dots(iv) \end{array}$$

Add Eq. (iv) and Eq. (iii)

$$\begin{array}{ll} (z + x) + (z - x) = 15 + 5 & \\ \Rightarrow 2z = 20 \Rightarrow z = 10 & \\ \text{from Eq. (iii), we have } 10 + x = 15 & \\ \Rightarrow x = 5 & \end{array}$$

and hence $y = 5 - x = 5 - 5 = 0$

$\therefore$ correct sequence is $z > x > y$

61. (c) Let present age of Ravi be x yr.

$\therefore$ Present age of Ravi's father = $4x$ yr

Now, 5 yr ago, age of Ravi's father

$$= (4x - 5) \text{ yr}$$

and age of Ravi = $(x - 5)$ yr

According to the question,

$$\begin{array}{ll} 4x - 5 = 7(x - 5) & \\ \Rightarrow 4x - 5 = 7x - 35 \Rightarrow 3x = 30 & \\ \therefore x = 10 & \end{array}$$

$\therefore$ Present age of Ravi = $x = 10$ yr

Present age of Ravi's father

$$= 4x = 4 \times 10 = 40 \text{ yr}$$

62. (d) Let the two positive numbers be x and y , respectively. According to the question,

$$\begin{array}{ll} (x + y) = 25(x - y) & \\ \Rightarrow x + y = 25x - 25y & \\ \Rightarrow 35y = 15x \Rightarrow \frac{x}{y} = \frac{7}{3} \text{ or } x = \frac{7}{3}y & \end{array}$$

Now, product of numbers $xy = 84$

$$\Rightarrow \frac{7}{3}y \times y = 84 \Rightarrow y^2 = \frac{84 \times 3}{7}$$

$$\Rightarrow y^2 = 12 \times 3$$

$$\therefore y = 6 \Rightarrow x = \frac{7}{3} \times 6 = 14$$

$$\begin{array}{ll} \therefore \text{Sum of numbers} = x + y & \\ = 14 + 6 = 20 & \end{array}$$

63. (a) Let the cost of one chair be ₹ x and that of one table be ₹ y .

$$\text{Then, } 2x + y = 700 \quad \dots(i)$$

$$\text{and } x + 2y = 800 \quad \dots(ii)$$

On multiplying Eq. (ii) by 2 and subtracting from Eq. (i), we get

$$\begin{array}{r} 2x + y = 700 \\ 2x + 4y = 1600 \\ \hline -3y = -900 \Rightarrow y = 300 \end{array}$$

On putting $y = 300$ in Eq. (ii), we get

$$\begin{array}{ll} x + 2 \times 300 = 800 & \\ x = 800 - 600 = 200 & \end{array}$$

Since, m chairs and m tables are to be purchased for ₹ 30000.

$\therefore$ Cost of m table and m chairs = 30000

$$\begin{array}{ll} \therefore 200m + 300m = 30000 & \\ \Rightarrow m = \frac{30000}{500} \Rightarrow m = 60 & \end{array}$$

64. (c) Let the two-digit number be $10x + y$,
Condition I

According to the question,

$$\begin{array}{ll} 10x + y = 3(x + y) & \\ \Rightarrow 10x + y - 3x - 3y = 0 & \\ \Rightarrow 7x - 2y = 0 & \dots(i) \end{array}$$

Condition II

$$\begin{array}{ll} \Rightarrow 10x + y + 45 - 10y - x = 0 & \\ \Rightarrow 9x - 9y + 45 = 0 & \\ \Rightarrow x - y = -5 & \dots(ii) \end{array}$$

On solving Eqs. (i) and (ii), we get

$$x = 2 \text{ and } y = 7$$

$\therefore$ Sum of the squares of digits

$$= (2)^2 + (7)^2 = 4 + 49 = 53$$

65. (a) Let x be the required number.

$$\begin{array}{ll} \therefore 52x - 25x = 324 & \\ \Rightarrow 27x = 324 \Rightarrow x = \frac{324}{27} = 12 & \end{array}$$

Hence, the required number is 12.

66. (b) Let a tin of oil contain x bottles of oil.

According to the question,

$$\begin{array}{ll} \frac{4}{5} - \frac{6}{x} + \frac{4}{x} = \frac{3}{4} \Rightarrow \frac{4}{5} - \frac{3}{4} = \frac{6}{x} - \frac{4}{x} & \\ \Rightarrow \frac{1}{20} = \frac{2}{x} \Rightarrow x = 40 & \end{array}$$

$\therefore$ A tin contains 40 bottles of oil.

67. (c) Let the number be $10x + y$.

$$\text{We have, } x + y = 7 \quad \dots(i)$$

$$\text{and } 10y + x = 10x + y + 27$$

$$\Rightarrow 9y - 9x = 27 \Rightarrow y - x = 3 \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get $y = 5$ and $x = 2$

$\therefore$ Number is 25.

$$\text{Product of number} = 2 \times 5 = 10$$

68. (c) Given, $\frac{p}{x} + \frac{q}{y} = m$ and $\frac{q}{x} + \frac{p}{y} = b$

$$\text{Let } \frac{1}{x} = u \text{ and } \frac{1}{y} = v$$

$$\text{Then, } pu + qv = m \quad \dots(i)$$

$$\text{and } qu + pv = n \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$u = \frac{mp - nq}{p^2 - q^2} \text{ and } v = \frac{mq - np}{q^2 - p^2}$$

$$\begin{array}{ll} \therefore \frac{x}{y} = \frac{1/u}{1/v} = \frac{v}{u} = \frac{-(mq - np)}{mp - nq} & \\ = \frac{np - mq}{mp - nq} & \end{array}$$

69. (c) Given equations are,

$$3x - ky - 20 = 0 \quad \dots(i)$$

$$6x - 10y + 40 = 0 \quad \dots(ii)$$

since, the given system has no solution

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{3}{6} = \frac{k}{10} \Rightarrow k = 5$$

70. (b) Let the ages of three brothers be a, b and c .

$$\text{Then, } a + b = 4, \quad b + c = 6$$

$$\text{and } c + a = 8$$

On solving these three equations, we get

$$a = 3, \quad b = 1 \text{ and } c = 5$$

$\therefore$ Age difference between eldest and youngest = $5 - 1 = 4$ yr

71. (a) Let annual incomes of two persons be $9x$ and $7x$ and expenses be $4y$ and $3y$, respectively.

Then, according to the question,

$$9x - 4y = 2000 \quad \dots(i)$$

$$\text{and } 7x - 3y = 2000 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$9x - 4y = 7x - 3y \Rightarrow y = 2x$$

On putting the value of y in Eq. (i), we get

$$9x - 8x = 2000 \Rightarrow x = ₹ 2000$$

$\therefore$ Difference between their annual incomes = $9x - 7x = 2x = ₹ 4000$

72. (c) Let the unit's place number be y and ten's place number be x .

Then, number = $10x + y$

Now, after interchanging the digits,

New number = $10y + x$ and sum of digits = $x + y$

According to the question,

$$10x + y = k(x + y) \quad \dots(i)$$

$$\text{and } 10y + x = m(x + y) \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$11(x + y) = (k + m)(x + y)$$

$$\Rightarrow k + m = 11 \Rightarrow m = 11 - k$$

18

QUADRATIC EQUATIONS AND INEQUALITIES

Regularly (9-10) questions have been asked from this chapter. Questions from this section generally focus on finding roots of quadratic equation and their factorisation.



QUADRATIC EQUATION

A quadratic equation is an equation whose degree is 2, meaning that the highest exponent of variable is 2. The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a , b and c are real numbers and $a \neq 0$.
e.g. $2x^2 + 3x + 5 = 0$, $x^2 + x + 1 = 0$ and $5x^2 + 6x + 8 = 0$ are all quadratic equations.

Roots of a Quadratic Equation

A real number α is said to be a root of the quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$), if it satisfies the equation i.e. $a\alpha^2 + b\alpha + c = 0$. We can also say that $x = \alpha$ is a solution of the quadratic equation.
e.g. 2 is the root of the quadratic equation, $x^2 - 6x + 8 = 0$ since $(2)^2 - 6(2) + 8 = 0$.

Note A quadratic equation has exactly two roots.

If α be the root of the quadratic equation $ax^2 + bx + c = 0$, then $(x - \alpha)$ is the factor of $ax^2 + bx + c$.

Solutions of Quadratic Equations

Solutions of quadratic equations can be determined by using any of the following two methods

1. By Factorisation

In this method, the middle term of the quadratic equation is splitted and the given quadratic equation is converted into a product of two linear factors. Then, the roots of the equation are obtained by equating each factor equal to zero.

Let the quadratic equation be $ax^2 + bx + c = 0$ and its linear factors are $(px + q)$ and $(rx + s)$, then

$$ax^2 + bx + c = (px + q)(rx + s)$$

$$\text{Now, } ax^2 + bx + c = 0 \Rightarrow (px + q)(rx + s) = 0 \Rightarrow px + q = 0 \text{ or } rx + s = 0 \Rightarrow x = \frac{-q}{p} \text{ or } x = \frac{-s}{r}$$

Thus, $\frac{-q}{p}$ and $\frac{-s}{r}$ are the roots of the equation, $ax^2 + bx + c = 0$.